

PROJECT PROPOSAL

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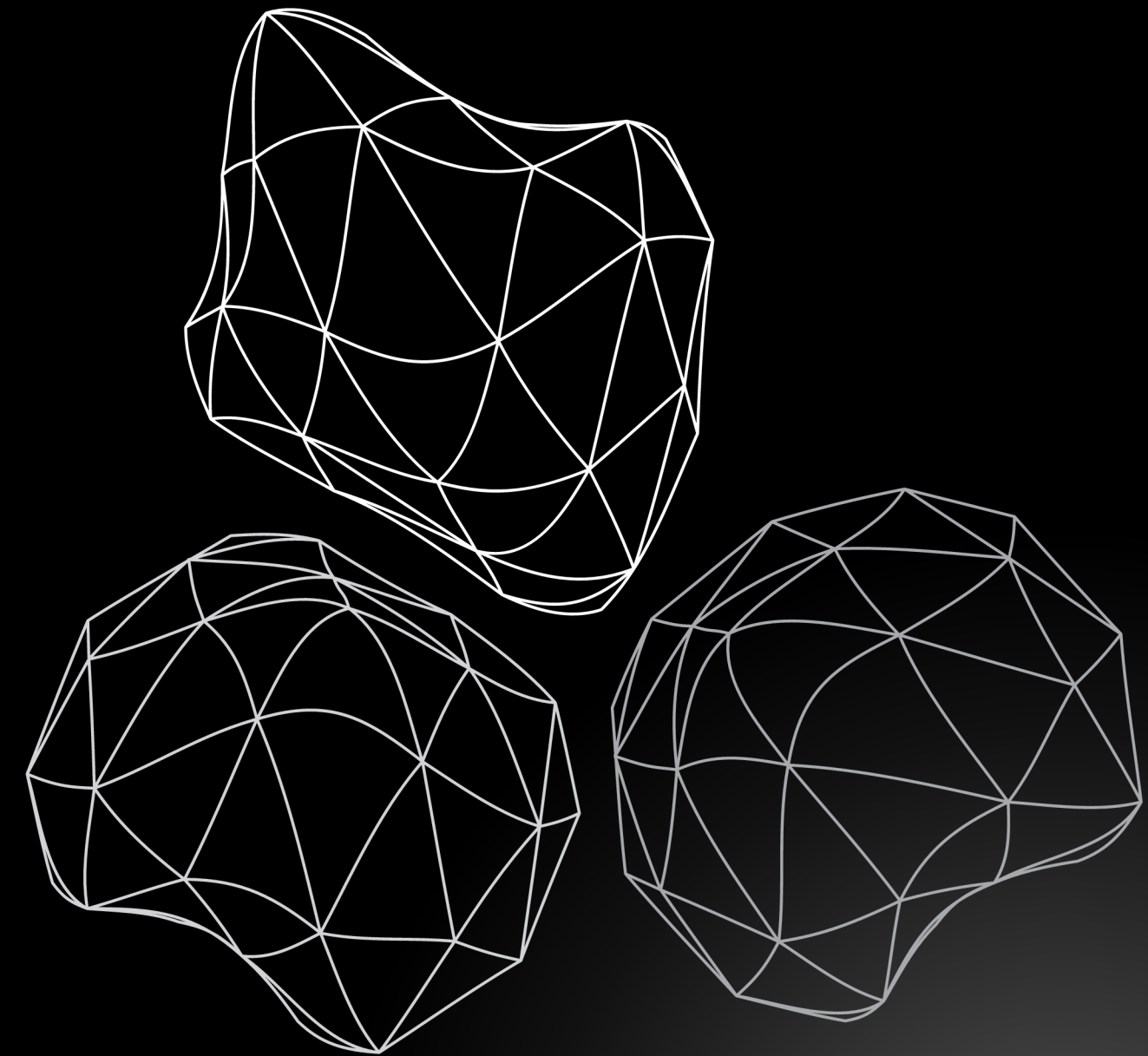
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RETAIL SOURCING AND SUPPLY COORDINATION SYSTEM

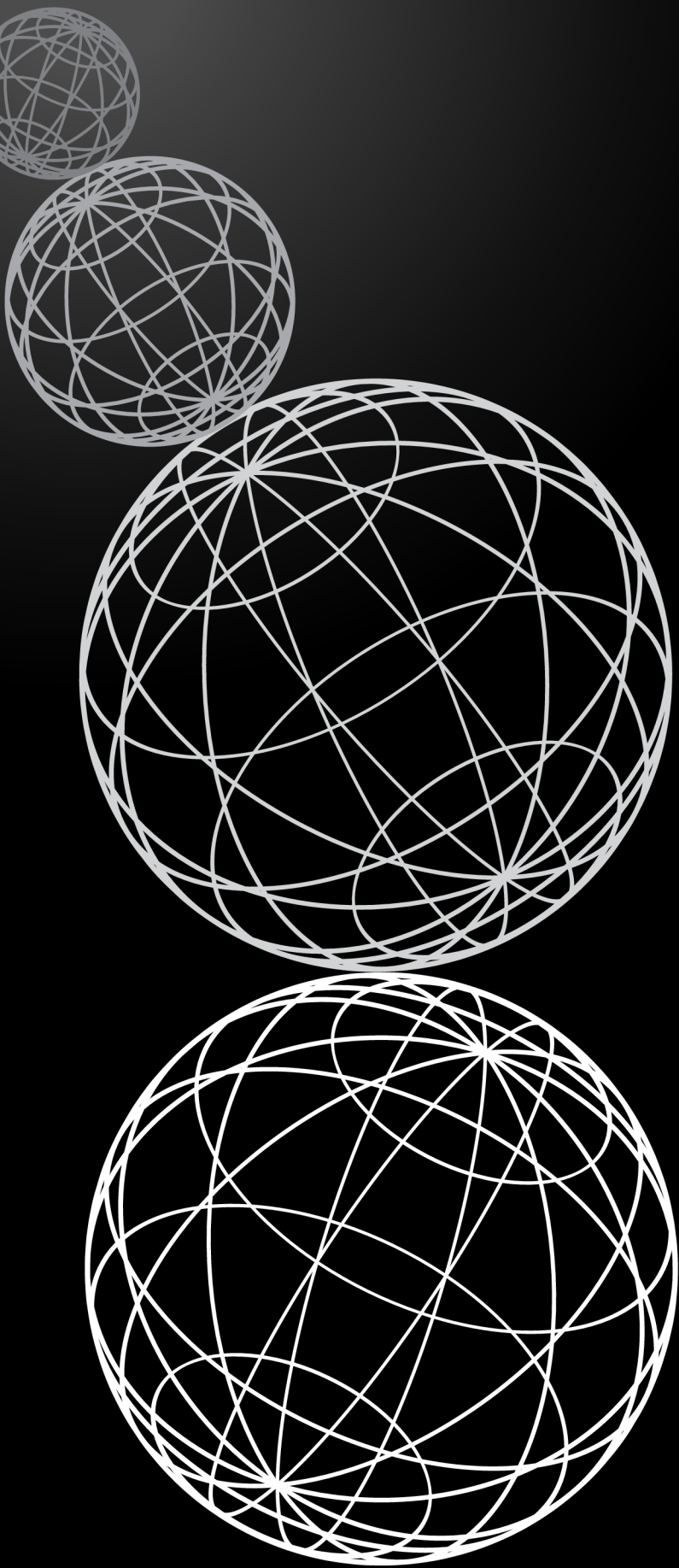
A Distributed Marketplace System for Retail Supply Chain Optimization

Retail Sourcing is a supply–demand coordination system designed to facilitate structured interaction between shop owners and product stockholders. The system enables shop owners to declare their product demand, including item details, quantity, and location, while allowing stockholders to publish available inventory information.

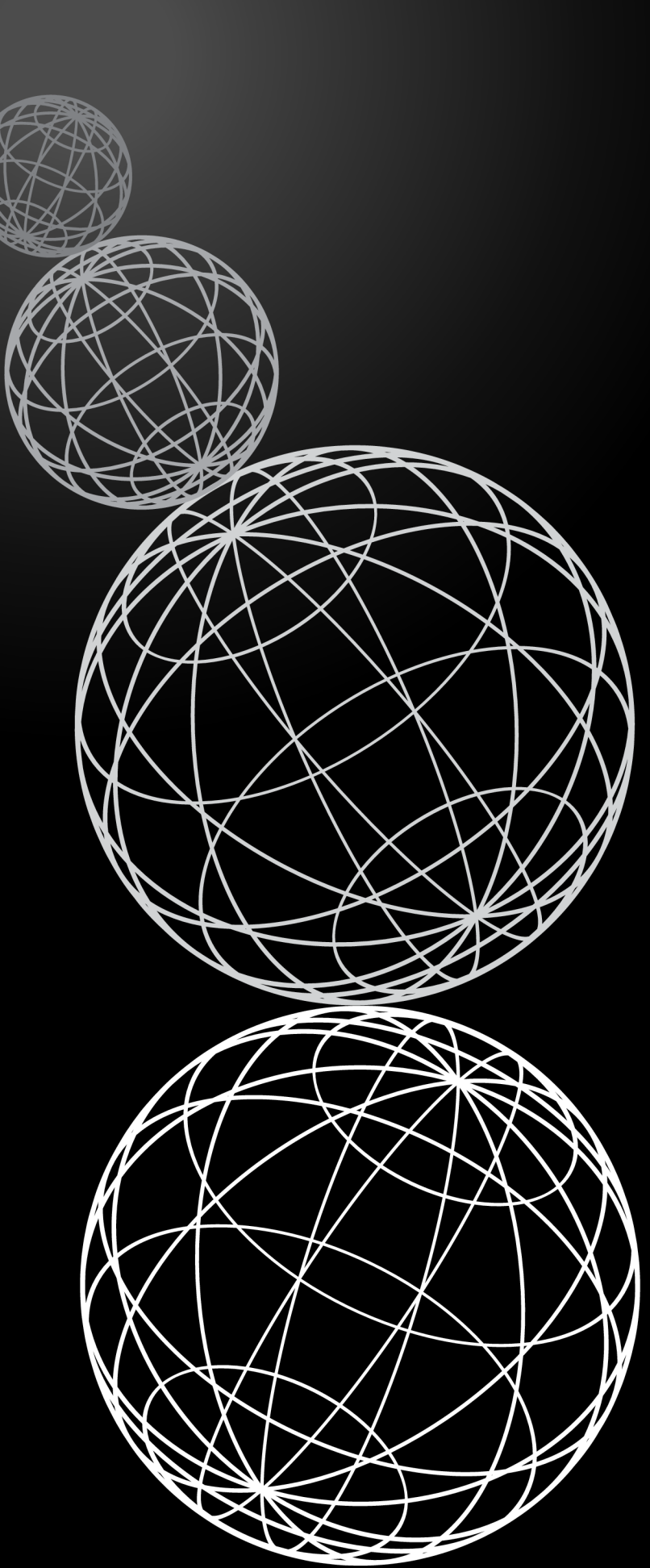


PROJECT CHARTER

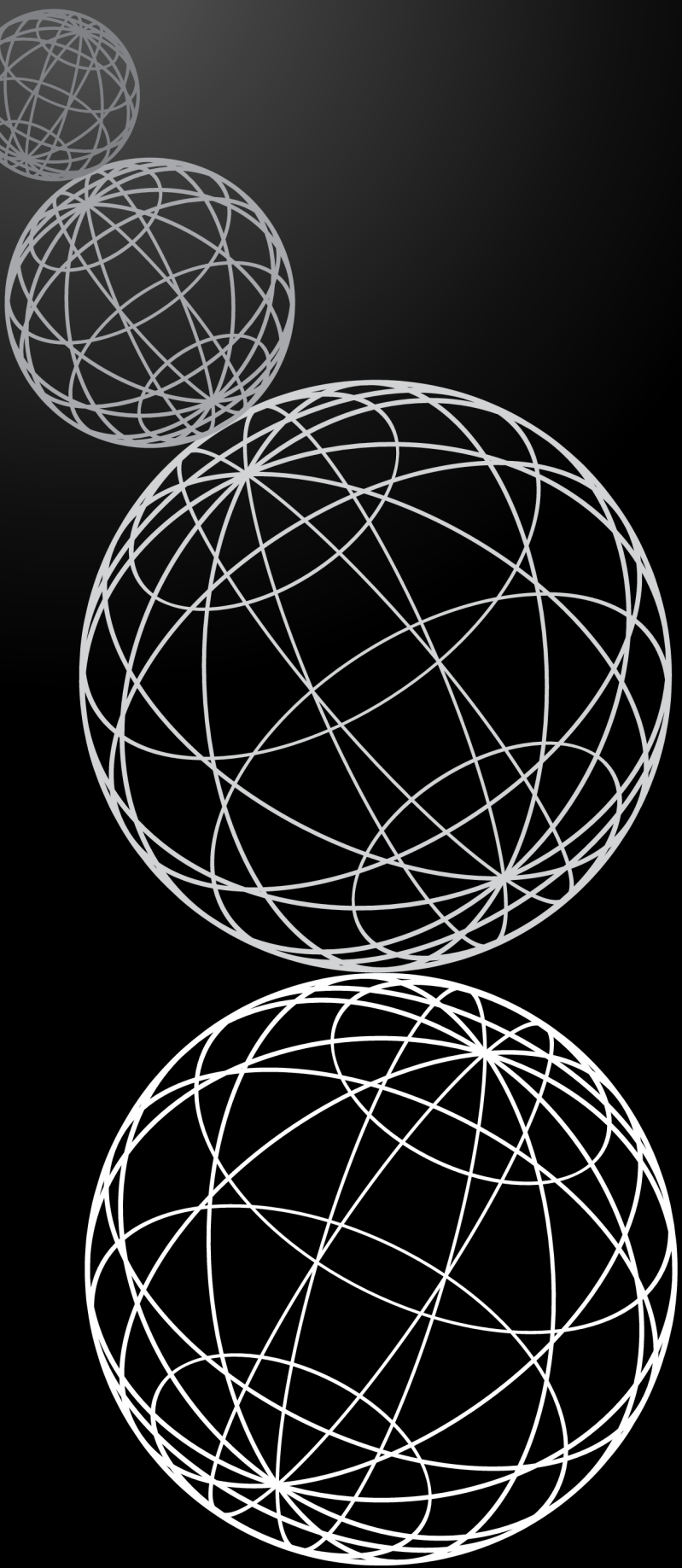
Section	Description
Problem Statement	Small retail shop owners and product suppliers rely on informal communication methods to connect for product sourcing. These methods lack real-time visibility of demand and stock availability, leading to sourcing delays, stock shortages, unsold inventory, and operational inefficiency. There is no centralized mechanism that enables structured and discoverable exchange of demand and supply information.
Vision statement	To establish a structured and efficient retail supply coordination environment where shop owners and suppliers can seamlessly discover and fulfill product demand in real time, minimizing delays and improving business efficiency.
Target users	1. Shop Owners (Buyers): Retail operators who need timely product sourcing. 2. Product Stockholders / Suppliers (Sellers): Inventory holders seeking buyers for their products.



PROJECT CHARTER



Section	Description
Objectives	1. Enable real-time visibility between buyers and suppliers. 2. Reduce sourcing delays and stock shortages. 3. Improve inventory utilization for suppliers. 4. Apply software engineering principles in system design and documentation.
Scope (In Scope)	- User registration and authentication - Demand posting by shop owners (product, quantity, location) - Stock posting by suppliers (product, quantity, availability) - Basic demand-supply visibility and filtering - Location-based search - Communication initiation between users
Scope (Out of Scope)	- Online payment integration - Delivery/logistics management - Advanced AI prediction systems - Financial accounting modules - International supply operations



PROJECT CHARTER

Section	Description
Risks	- Low user adoption - Inaccurate or outdated data entry - Scalability issues with growing users - Location data inaccuracies - Scope creep during development

SAMPLE PROBLEM & SOLUTION TABLE

Section	Description
Sample Problem	A shop owner urgently needs 50 units of a product but has no structured way to identify nearby suppliers. After multiple calls and delays, the product is sourced late, causing lost sales.
Sample Solution	The system allows the shop owner to post demand instantly. Suppliers with available stock can view and respond in real time, reducing delay and preventing lost sales.

FUNCTIONAL REQUIREMENTS

The system shall allow users to register as either a shop owner or a product stockholder.

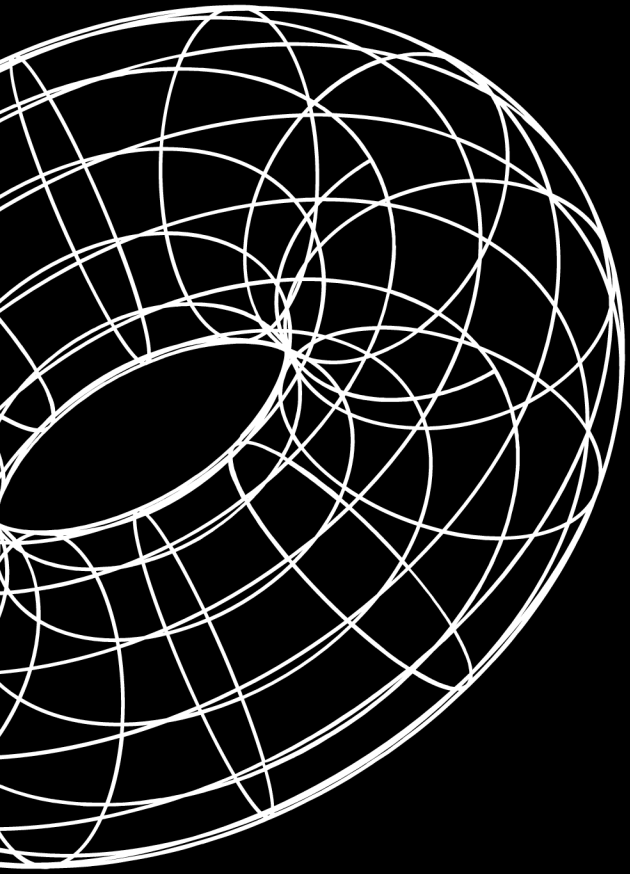
The system shall authenticate users using secure login credentials.

The system shall allow shop owners to submit product demand information including product name, quantity, and location.

The system shall allow users to register as either a shop owner or a product stockholder.

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NON-FUNCTIONAL REQUIREMENTS

(Performance)

The system shall respond to user requests within 2 seconds under normal load conditions (up to 500 concurrent users).

(Scalability)

The system shall support at least 5,000 registered users without performance degradation.

(Availability)

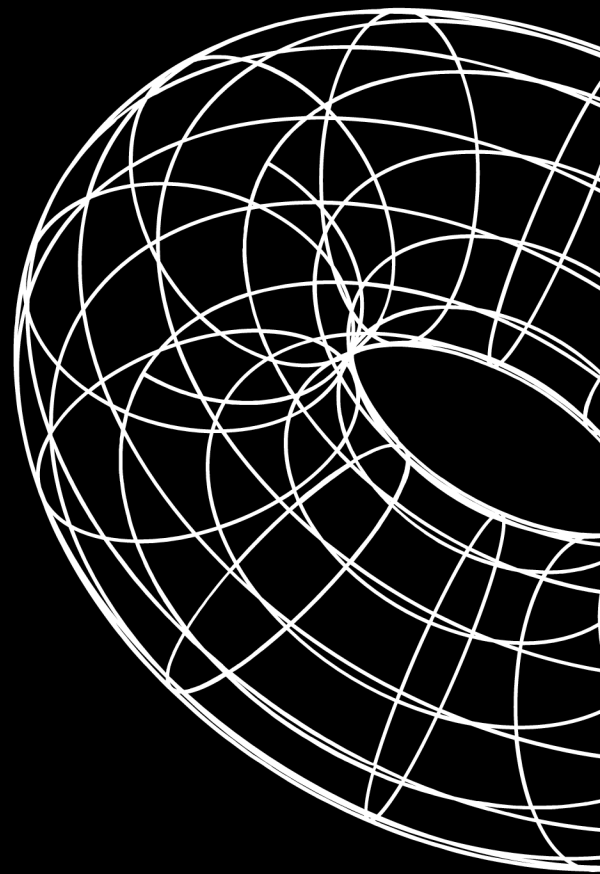
The system shall maintain 99% uptime excluding scheduled maintenance.

(Security)

The system shall encrypt user authentication credentials and sensitive data using industry-standard encryption protocols.

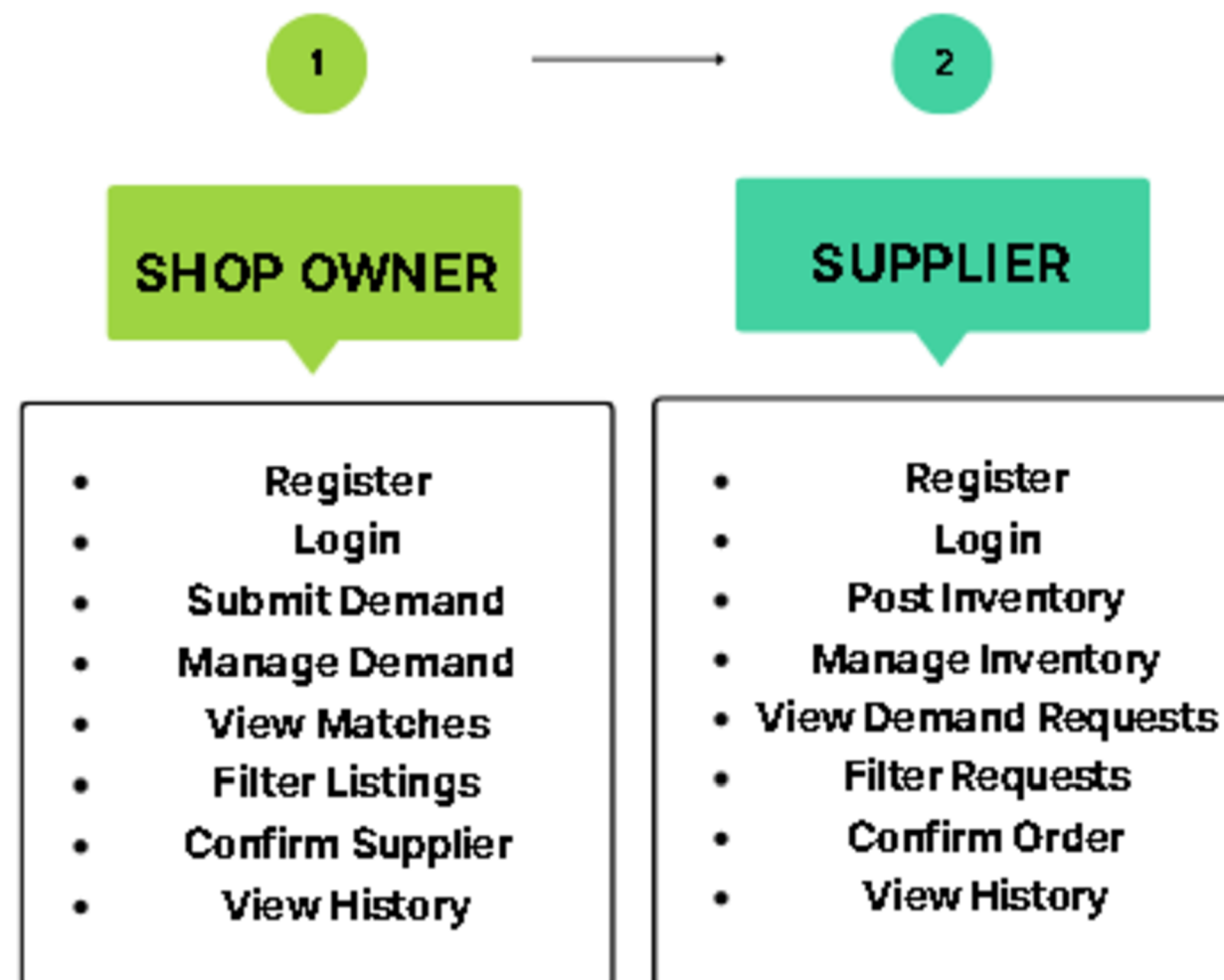
(Usability)

The system shall allow a new user to complete registration and submit a demand or stock entry within 3 minutes without external assistance.



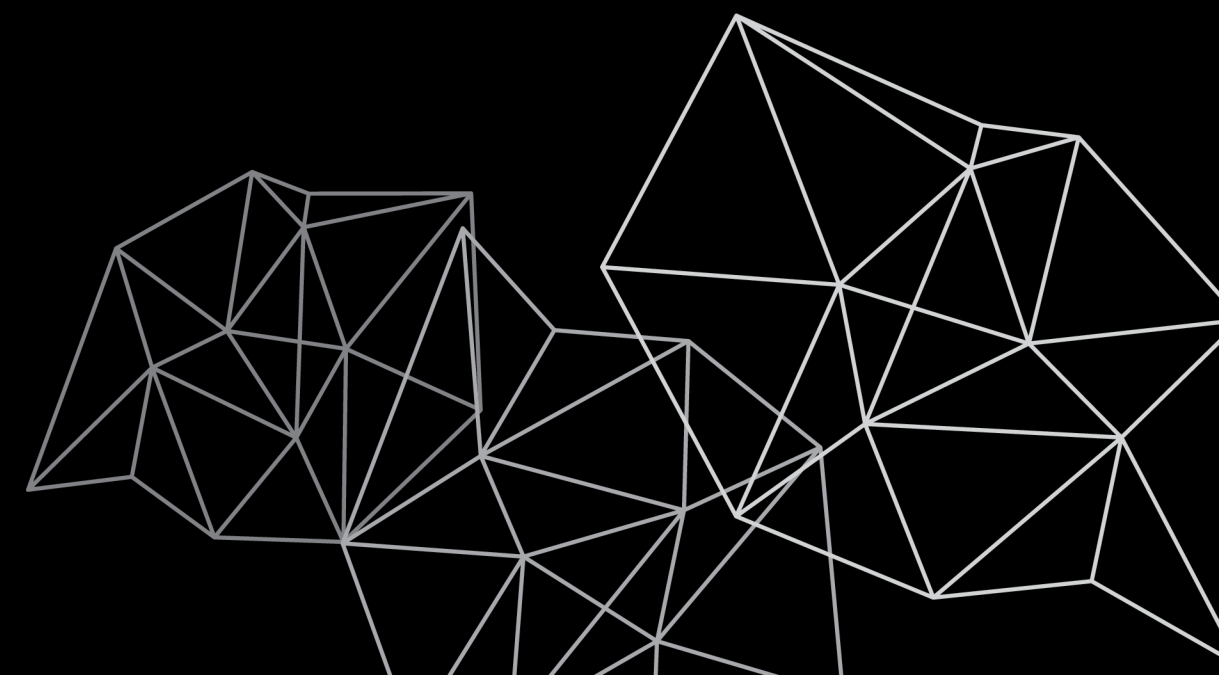
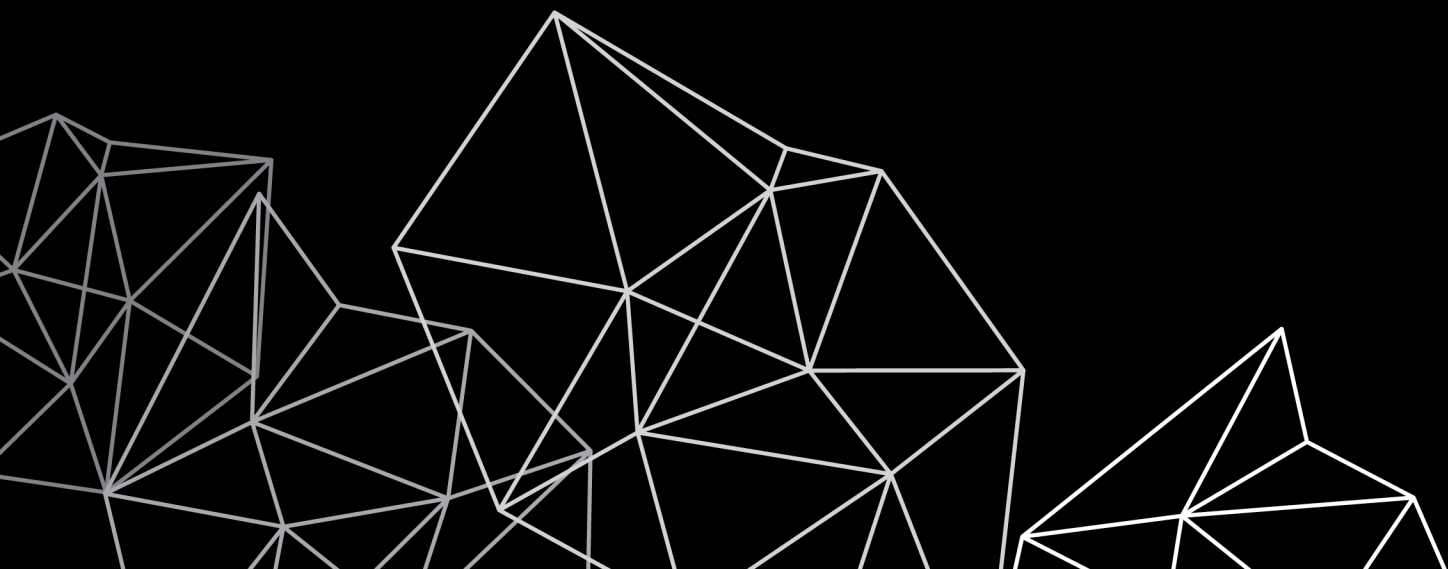
DIAGRAM

Retail Sourcing System



MARKET OVERVIEW

- As a shop owner, I want to submit my product demand with quantity and location, so that suppliers who have the required stock can identify and respond to my request quickly.
- As a shop owner, I want to view available suppliers who have matching products, so that I can source items on time and avoid stock shortages.
- As a supplier, I want to publish my available inventory details, so that shop owners who need those products can discover and purchase them.
- As a supplier, I want to view active product demand requests from nearby shop owners, so that I can sell my stock efficiently and reduce unsold inventory.



USER STORIES WITH ACCEPTANCE CRITERIA

PROJECT: RETAIL SOURCING SYSTEM

As a shop owner, I want to submit my product demand with quantity and location, so that suppliers can identify and respond to my request quickly.

- The system shall allow a shop owner to submit a demand, including product name, quantity, and location.
- The system shall reject the submission if any required field (product name, quantity, or location) is missing.
- The system shall store the submitted demand and make it visible to relevant suppliers.

As a shop owner, I want to view available suppliers who have matching products, so that I can source items on time.

- The system shall display a list of suppliers whose inventory matches the requested product name.
- The system shall allow filtering results based on location.
- The system shall update the supplier list dynamically when new matching inventory is posted.

As a supplier, I want to publish my available inventory details, so that shop owners can discover and purchase my products.

- The system shall allow a supplier to submit inventory information, including product name and quantity.
- The system shall allow the supplier to update or remove inventory entries.
- The system shall make the published inventory visible to shop owners searching for matching products.

High-Level Software Architecture Diagram

SHOP OWNER (BUYER)

- Submit Demand
- View Matching Suppliers
- Filter Listings
- View Interaction History

RETAIL SOURCING SYSTEM

- Demand Management Module
- Supply Management Module
- Matching & Notification
- Filtering Engine
- Interaction Logging

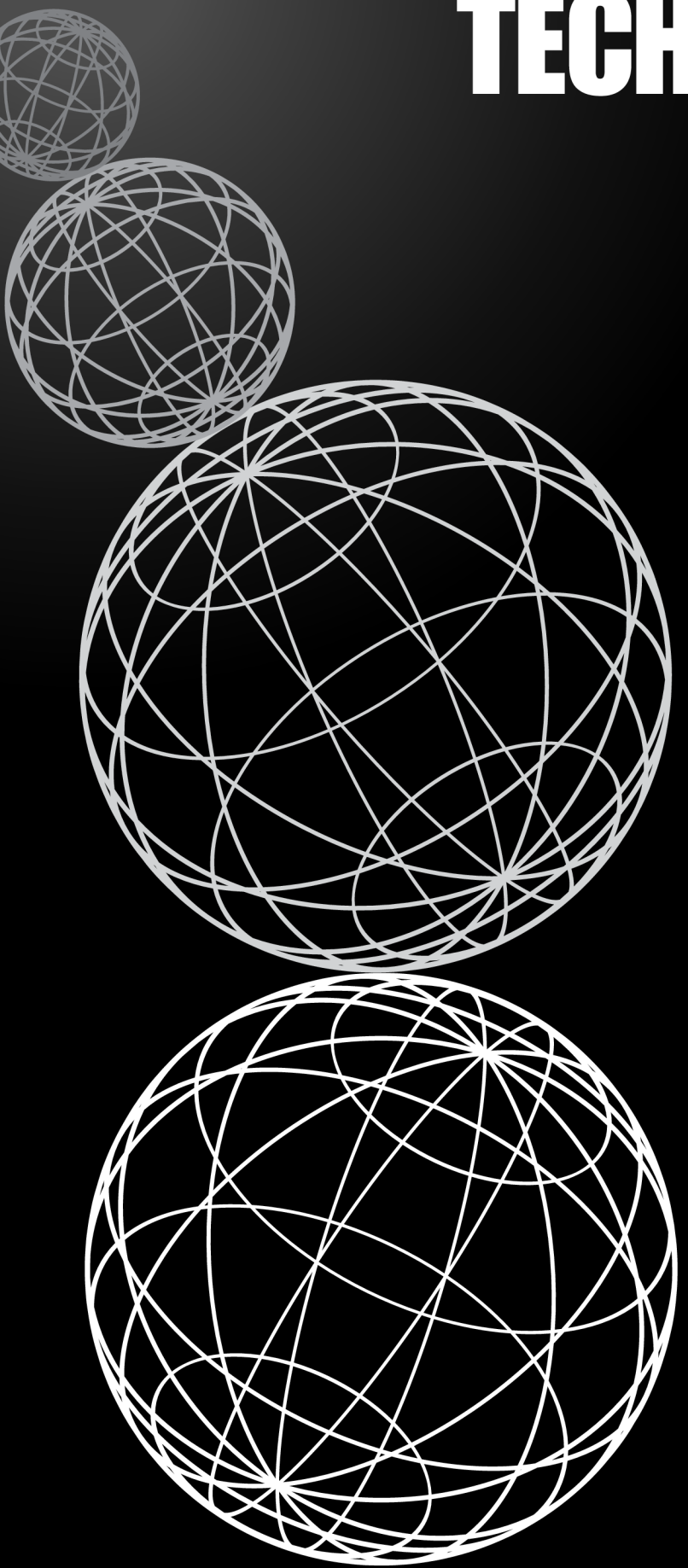
PRODUCT STOCKHOLDER
(SUPPLIER)

- Publish Inventory
- View Active Demands
- Filter Demands
- Interaction History

DATABASE / STORAGE

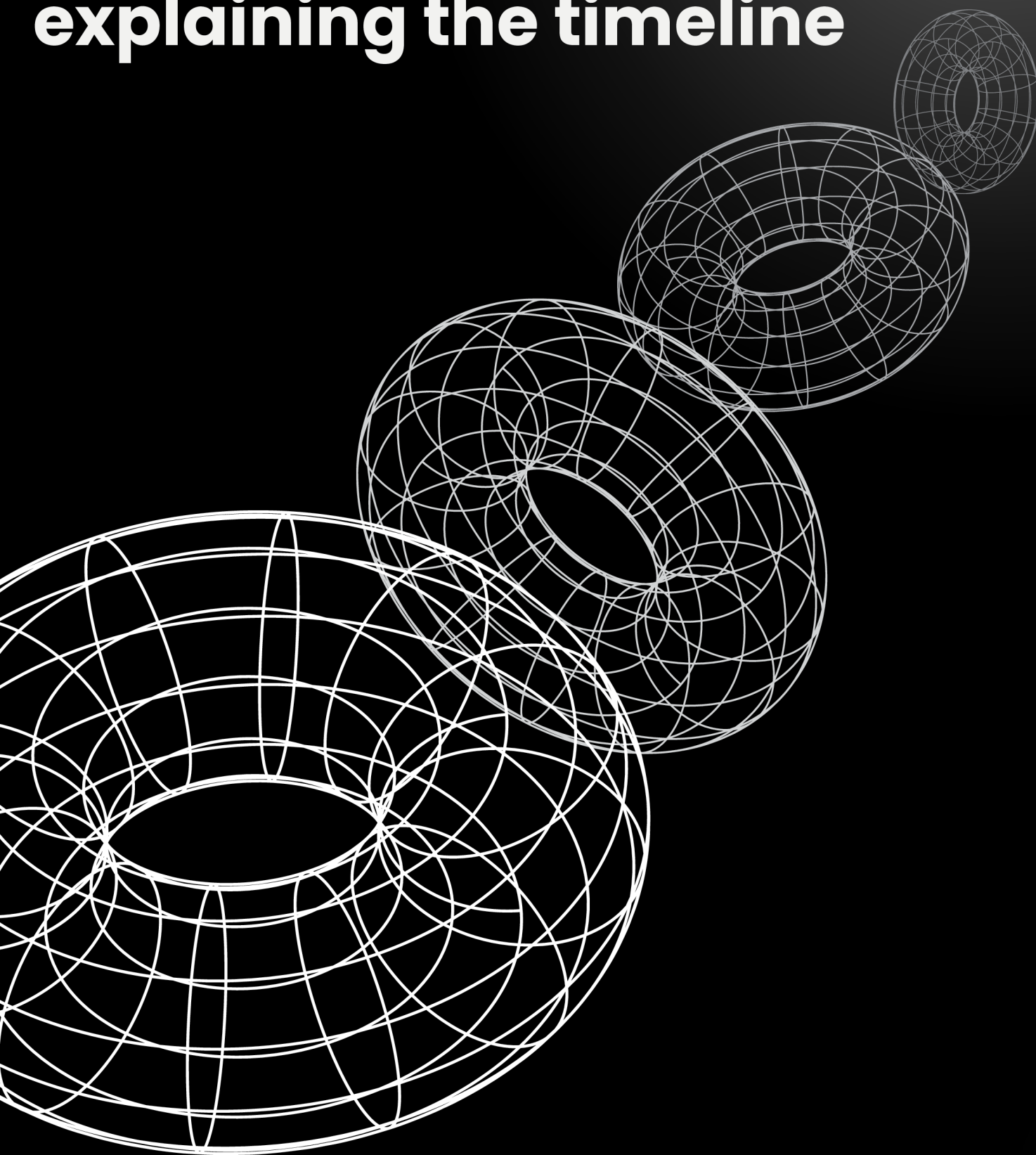
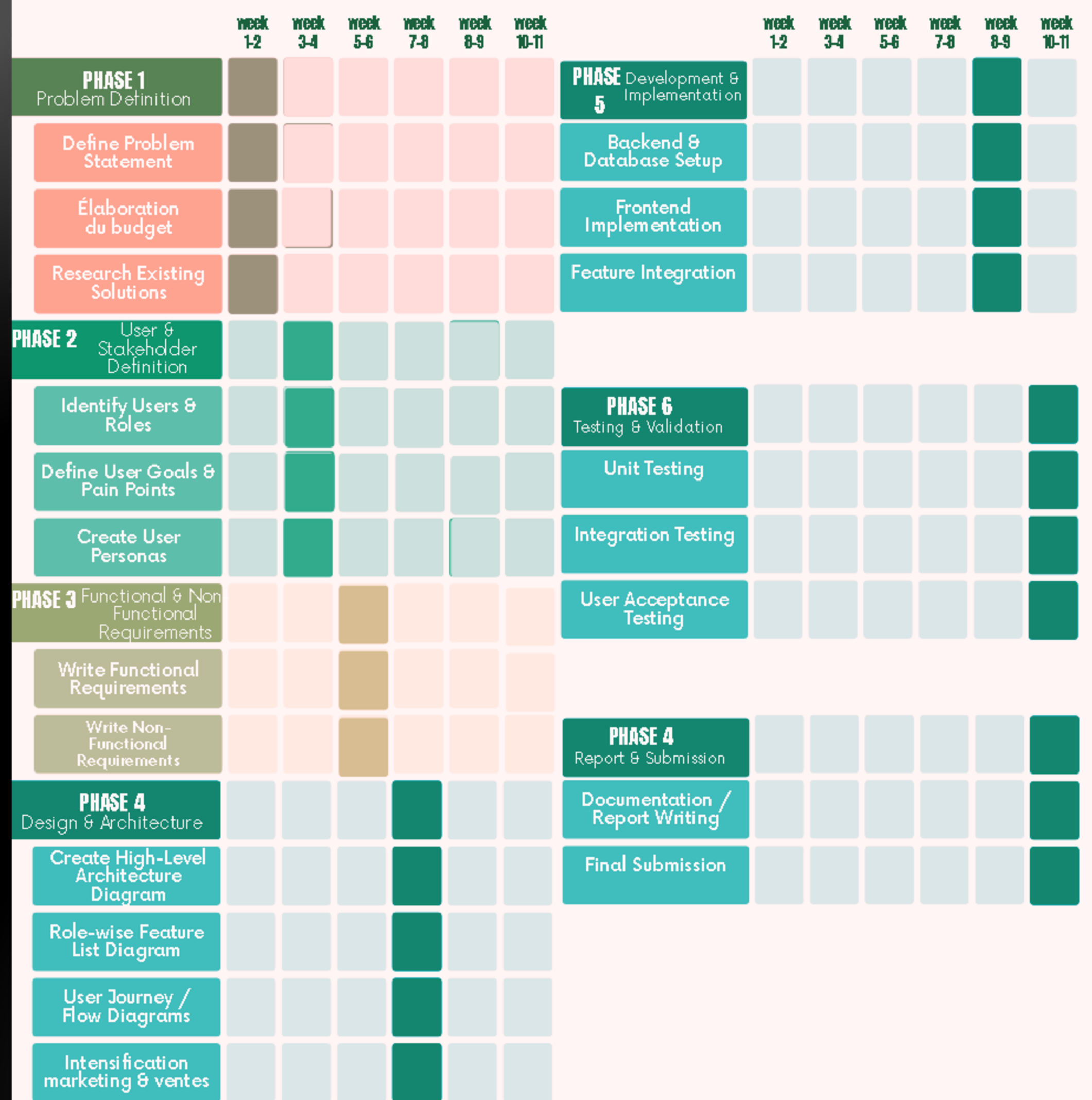
- Demand Records
- Inventory Records
- User Records
- Interaction Logs

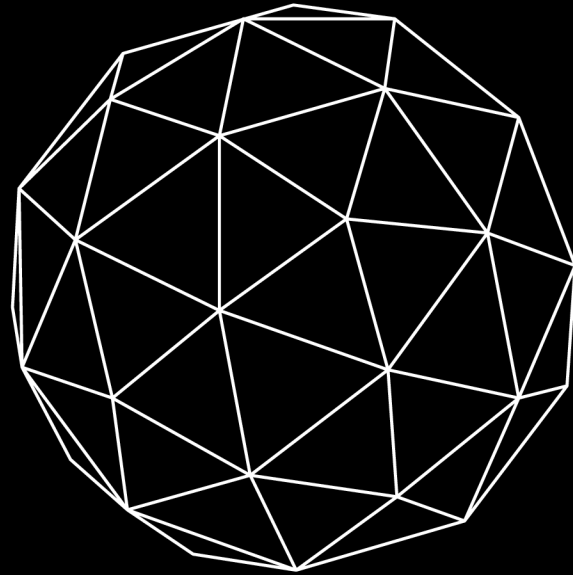
TECHNOLOGY STACK FOR RETAIL SOURCING SYSTEM



Layer	Technology / Tool	Purpose / Justification
Frontend (UI / Clients)	Flutter	Cross-platform mobile & web interface, allows rapid prototyping for shop owners and suppliers
Backend (Server / API)	flask- REST API	Handles requests, processes business logic, scalable and easy for academic projects
Database	mysql	Relational DB for storing users, inventory, demand, and interactions; supports queries & filtering
Hosting/ Deployment	Firebase	Cloud deployment for database and backend services; simple setup for academic project

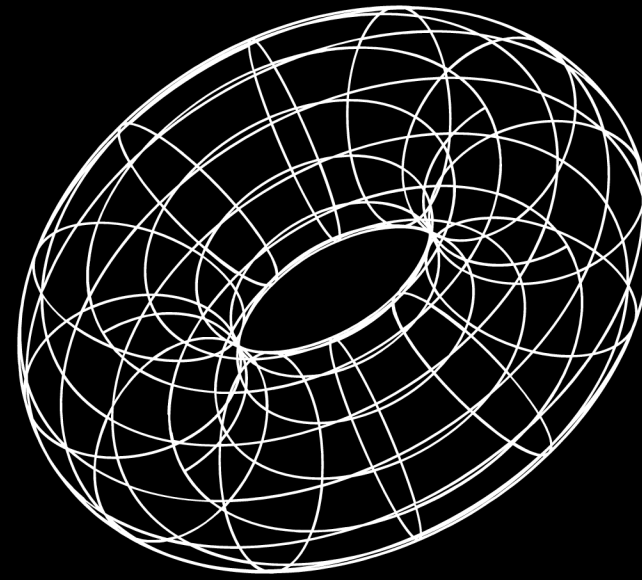
explaining the timeline

The image features four white wireframe tori (donut shapes) on a black background, arranged in a diagonal sequence from the bottom-left towards the top-right. Each torus is composed of a grid of intersecting circles. The tori increase in size and opacity from the smallest and faintest at the top-right to the largest and most prominent at the bottom-left.



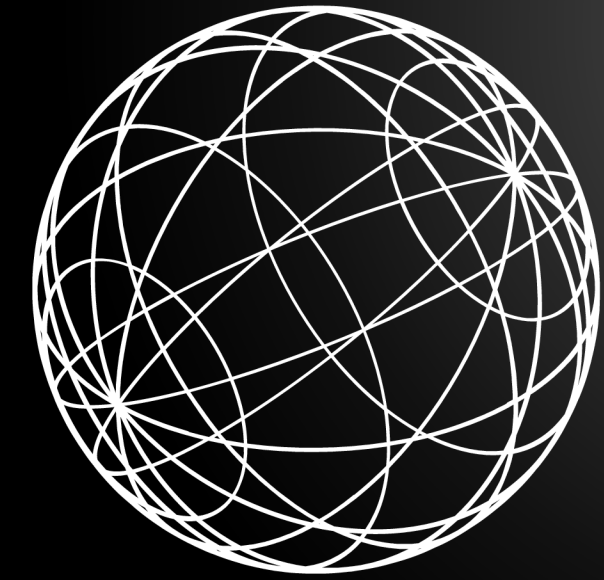
Member-1

- Lead problem statement drafting
- Define Shop Owner user type
- Create Shop Owner persona
- Lead Functional Requirements drafting
- Design High-Level System Architecture
- Set up Backend & Database
- Conduct Unit Testing
- Draft report sections (Introduction, Problem, Objectives)



Member-2

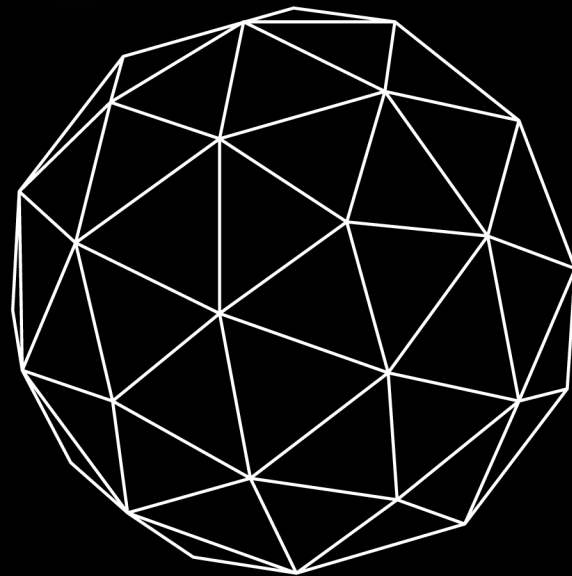
- Research existing solutions
- Define Supplier user type
- Create Supplier persona
- Lead Non-Functional Requirements drafting
- Create Role-wise Feature List Diagram
- Implement Frontend
- Conduct Integration Testing
- Draft report sections (Requirements, Design)



Member-3

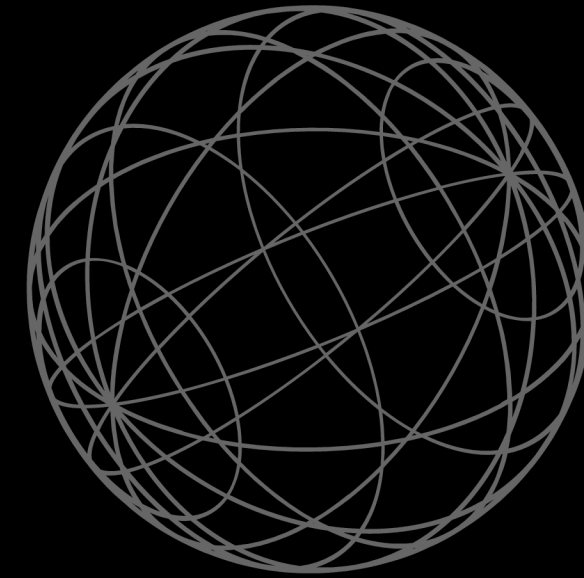
- Review & edit problem draft
- Identify stakeholders
- Compile personas
- Review & verify testability of all requirements
- Create User Journey / Flow Diagrams
- Handle Feature Integration & Testing
- Conduct User Acceptance Testing (UAT)
- Draft report sections (Testing, Conclusion, Appendices)

REFERENCES (2 SIMILAR PRODUCTS)



1. Alibaba
2. IndiaMart

proposed novelty



- Demand-driven (reverse marketplace) model where shop owners post requirements first
- Location-based smart matching between shop owners and nearby stockholders
- Real-time inventory visibility for accurate availability tracking
- Role-based dashboards (Shop Owner, Stockholder, Admin)
- Focused specifically on small and local retailers
- Built-in structured negotiation and confirmation workflow
- Demand analytics to help stockholders predict high-demand products
- Reduced middlemen through direct shop owner–stockholder coordination

THANK YOU